

Ojas Srivastava

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Education

Bachelor of Technology in Artificial Intelligence

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India

CGPA: 9.20/10

Aug 2024 – May 2028

- Expected graduation **May 2028**; available for summer internship **May–Aug 2027**
- Coursework:** Probability & Statistics, Linear Algebra, Machine Learning, Data Science, Deep Learning, Design & Analysis of Algorithms, Discrete Mathematics, DBMS/**SQL**, Artificial Intelligence, Optimization Techniques

Senior Secondary (Class XII), CBSE

Shiv Jyoti Convent School, Kota, India

95.80%

2024

Secondary Education (Class X), ICSE

St. Joseph's College, Prayagraj, India

97.20%

2022

Professional Experience

Software Engineering Intern — Data-Aware Production Systems

IFFCO (Indian Farmers Fertiliser Cooperative), Phulpur Unit, Prayagraj, India

Jun 2025 – Jul

2025

- Queried and prepared operational **MySQL/SQL** datasets for reporting and automation (**50+** **daily** workflows); reviewed data quality (missing fields, duplicates, validation rules) before production use.
- Built **10+ RESTful APIs** (Node.js/Express) with structured error handling and automated **pytest**-style validation; presented analytical findings and trade-offs to cross-functional stakeholders.
- Managed parallel workstreams (development, testing, documentation, stakeholder updates); learned new systems quickly and adjusted to shifting priorities.

Quantitative & ML Projects

Options Pricing Model — Monte Carlo & Black-Scholes Engine

C++17, Stochastic Modeling, Geometric Brownian Motion, Greeks, Risk-Neutral Pricing

2026

GitHub

- Engineered Monte Carlo simulation engine in **C++** for European options pricing using **Geometric Brownian Motion** with **100,000** simulated paths, Mersenne Twister PRNG, and risk-neutral discounting.
- Implemented Black-Scholes analytical pricer computing full **Greeks** (Delta, Gamma, Theta, Vega, Rho), **implied volatility**, and intrinsic value — comprehensive risk assessment for derivatives valuation.

LogiFlow — Predictive Modeling & Multi-Objective Optimization (Google Solution Challenge 2026, Global Top 106)

Python, Scikit-learn, Pandas, NumPy, SQL, FastAPI, Redis, Cross-Validation

Apr 2026 – Jun 2026

GitHub | Live

- Trained **HistGradientBoosting** and **GradientBoosting** regressors on **15,650+** train-day records with **13 engineered features** (cyclical hour encoding, log-distance transform, junction flags); achieved **MAE ≤ 25 min**, **81%** within 30-min window.
- Designed **GroupKFold** cross-validation (grouped by train number to prevent leakage) and **leave-one-date-out backtesting**; auto-retuned model when R^2 fell below 0.35.
- Built **priority-weighted multi-objective optimizer** (cost/time/risk Pareto ranking) with cargo-aware dynamic weight adjustments and composite risk scoring from 5 weighted components (real delays, seasonal, weather, transfer, train-type).
- Wrote **SQL/Python** ETL pipelines ingesting RailRadar API, scraped delay corpus, and weather data; validated with **100/100** automated business-rule checks.

AirHelp — Retrieval & Decision-Support System (PowerMind Hackathon 2026)

Python, NumPy, Embeddings/RAG, FastAPI, WebSockets

2026

GitHub

- Built semantic retrieval system over structured + unstructured knowledge with embedding-based indexing; evaluated quality under **50–100** concurrent users (latency **1–3s**).

Positions of Responsibility

- Chairperson, Nexus SVNIT (2026–Present):** Head the departmental technical cell of SVNIT; organize workshops, coding contests, and mentoring programs for **500+** students across the department.
- Executive Member, ACM SVNIT:** Led competitive programming workshops and DSA boot-camps; assertive facilitation and peer mentoring.
- Technical Co-Lead, LogiFlow (Global Top 106):** Coordinated ML modeling, ETL, and deployment workstreams under a fixed deadline; presented to judges.
- McKinsey.org Forward Fellow (2026):** Structured problem-solving, analytical communication, and professional judgment program.

Achievements

- Machine Learning Specialization** (Andrew Ng, Stanford / DeepLearning.AI, Coursera, Mar 2026) — supervised learning, neural networks, unsupervised learning, recommender systems, reinforcement learning [Verify]
- LeetCode Knight:** peak rating **2048**, **744** problems solved, 32 contests; **Codeforces Specialist:** rating **1419**, **265** solved — strong analytical and algorithmic problem-solving
- Google Solution Challenge 2026** — **Global Top 106** (LogiFlow); **Vibe2Ship 2026** — **Global Top 20** (Community Hero)

Technical Skills

Languages: Python, C/C++, **SQL**, Java, TypeScript/JavaScript

Quantitative / ML: Statistical modeling, **Scikit-learn**, Pandas, NumPy, feature engineering, cross-validation (GroupKFold, stratified), Gradient Boosting, stochastic simulation, derivatives pricing, Greeks

Data & Infra: MySQL, Firebase/Firestore (NoSQL), Redis, FastAPI, REST APIs, ETL pipelines, **Big Data** awareness (Spark concepts)

Tools: Git, Docker, Linux, pytest, CI/CD (GitHub Actions), Excel

Communication: Good spoken and written English; effective at presenting technical concepts to non-technical stakeholders

Approach: Strong logical, analytical skills and a systematic problem-solving approach